



Frederico José Ribeiro Pelogia

Computer Engineering
MSc. Biomedical Engineering

Contact Information

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Portfolio (GitHub):

<https://github.com/fpelogia/fpelogia.github.io>

Location:

São José dos Campos - SP
Brazil

Languages

English - Fluent

Skills

Programming Languages:

- Python
- Julia
- C / C++
- MATLAB/Simulink
- JS/TS/Node

Others

- Mathematics for Engineering & Data Science
- Problem Solving / Modelling
- Full-Stack Software Development
- Communication & Collaboration

Published Papers

Multi-wave modelling and short-term prediction of ICU bed occupancy by patients with Covid-19 in regions of Italy
Mathematical Modelling of Natural Phenomena 19, 13 - (2024)

Short-term prediction of COVID-19 deaths in Argentina
IFMBE (International Federation for Medical and Biological Engineering) Proceedings - CLAIB 2022

Study of the COVID-19 pandemic trending behavior in Israeli cities
IFAC Papers Online - 11th IFAC Symposium on Biological and Medical Systems BMS 2021

COVID-19 Trend Analysis in Mexican States and Cities
43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) 2021

A support tool for planning classrooms considering social distancing between students
Computational and Applied Mathematics (2022)

Objectives

Computer Engineer and MSc. in Biomedical Engineering specializing in mathematical and computational modeling. Leveraging a strong background in aeronautical data analytics, reliability engineering, and software development, I am seeking positions in Engineering, R&D, and Data Science, with a core focus on systems modeling, simulation, and control systems.

Experience



Embraer - Mtx Costs & Reliability Engineer

2021 - Now
(2025-2026)

- Mathematical modeling of the Direct Maintenance Costs (DMC) of the Executive Aviation Fleet
- Statistical analysis of component reliability, maintenance planning, costs and prices
- Studying aircraft systems for supporting business decisions

Embraer - Software Developer II - EVE

(2023-2025)

- Development of IT solutions for EVE Air Mobility's Support and Services division
- Full Stack development of web applications - Backend APIs (NodeJS/Python), Front-end (Angular), Database (PostgreSQL), Deploy with CI/CD Azure DevOps and AWS, Docker and K8S
- Collaborative planning of the data ecosystem and application architecture
- Working together with the business area (Reliability and Maintenance Costs), with active participation in solution planning
- Automating data collection and treatment processes (Python, VBA)
- Planning solutions and new projects in data science (NLP, SVM, OCR)

Embraer - IT Internship - Support & Services

(2021-2022)

- Support internal applications and APIs (Full-Stack - JS / Node) for Executive Aviation
- Automating processes with Python



MSc. Research Project - Mathematical Modelling

2021 - 2024

- Mathematical and computational modeling of the evolution of COVID-19 indicators, performing short-term forecasts of ICU bed occupancy by patients with the disease
- Model developed and implemented in Python, using libraries such as SciPy, Numpy, Matplotlib,
- Graphical interface to aid experimentation with the simulations using Streamlit (Python framework).



Undergraduate Research (FAPESP)

2019 - 2020

- "From scratch" implementation of deep neural networks using different optimization methods (GD, SGD, RMS Prop, Adam, Levenberg-Marquardt) applied to Fraud Detection.
- Collaboration with researchers in applied mathematics

Education



Federal University of São Paulo (UNIFESP)

2022 - 2024

Master's Degree in Biomedical Engineering

2018 - 2022

Computer Engineering

2018 - 2021

Interdisciplinary Bachelor in Science and Technology